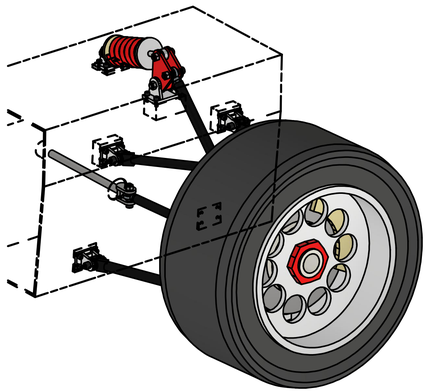
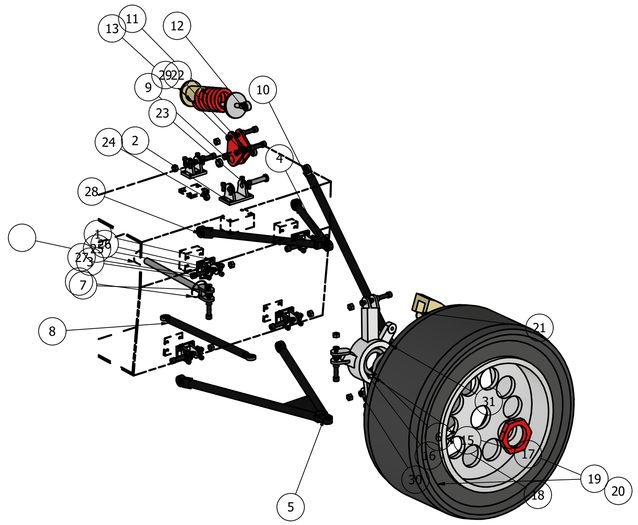
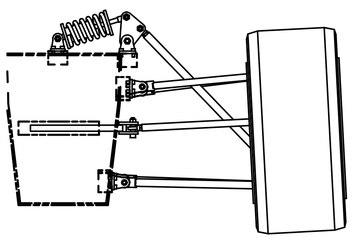
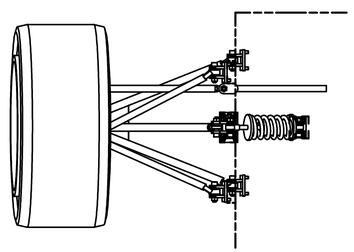




PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
17	1	F1S-404	Brake disc iron 320 x 32, ventilated, floating on 12 bobbins (vanes not modelled)
18	1	F1S-406	Wheel rim 18 in x 12 in forged aluminium, centre lock (envelope)
19	1	F1S-407	Front slick 30/68-18 (680 mm OD, unloaded envelope)
20	1	F1S-408	Centre-lock wheel nut M80x1.5, 7075-T6
21	1	F1S-405	Brake caliper, 6-piston monobloc for a 320 x 32 disc (envelope), 2x M10 mount
22	2	F1S-524	Spherical plain bearing SKF GE 15 ES (15 x 26 x 12, C0 85 kN), rocker pivot (x2)
23	1	F1S-515	Rocker pivot pin d15 h6 x 62, M12 thread, 4340 steel
24	13	F1S-523	All-metal prevailing-torque locknut M12, class 10
25	8	F1S-511	Spherical plain bearing GE 12 E - inner ring (d12, dk18, B10), catalogue dimensions
26	16	F1S-519	High-misalignment cone spacer 12.2 / 14-19 x 7, 17-4PH
27	12	F1S-521	Socket head cap screw ISO 4762 M12 x 55, class 12.9, black oxide
28	8	F1S-512	Spherical plain bearing GE 12 E - outer ring (D22, C7), pressed in housing
29	4	F1S-514	Rod end M12 - ball (bore 12, sphere 22, B12)
30	8	F1S-518	High-misalignment cone spacer 12.2 / 14-19 x 6, 17-4PH
31	4	F1S-513	Rod end male M12x1.25 RH body (Fluro steel, C0 42 kN; h 50, C9, D 32) with jam nut

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	5	F1S-101	Chassis clevis bracket, 7075-T6 machined, M12 double shear, 4x M8 to monocoque inserts
2	1	F1S-102	Rocker mount, 7075-T6 machined, d15 pivot pin double shear, 4x M8 to monocoque
3	24	F1S-522	Socket head cap screw ISO 4762 M8 x 20, class 12.9 (bracket to monocoque insert)
4	1	F1S-201	Upper wishbone, 4130 tube 22.2 x 1.65 TIG-welded, GE12E housings, normalised after welding
5	1	F1S-202	Lower wishbone, 4130 tube 25.4 x 1.65 TIG-welded, 3 mm gusset, GE12E housings (LBJ bolt vertical)
6	1	F1S-203	Front upright, 7075-T6 machined from billet: hub-bearing boss, UBJ/LBJ/steering/pushrod clevises, caliper lugs
7	1	F1S-104	Steering rack bar end with M12 tie-rod clevis, 17-4PH (rack teeth not modelled)
8	1	F1S-205	Tie rod (track rod), 4130 tube 22.2 x 1.65, M12 inserts, eye-to-eye 470.1 mm
9	1	F1S-206	Rocker (bell crank), twin 7075-T6 plates + hub, arms 60/80 mm, 2x SKF GE 15 ES (hub wall 10 mm)
10	1	F1S-204	Pushrod, 4130 tube 25.4 x 2.11, M12 RH/LH inserts, eye-to-eye 653.1 mm
11	1	F1S-301	Damper Penske 8760 triple-adjustable, 45 mm bore, 2.25 in spring hardware, built to 241.2 mm static / 262.5 mm extended eye-to-eye, stroke 63.5 mm (envelope)
12	1	F1S-302	Damper shaft + eye + lower spring perch (envelope)
13	1	F1S-303	Coil spring Hyperco 186A0900: 2.25 in ID, 6.0 in free, 900 lb/in (157.6 N/mm), installed 142.2 mm (envelope: wire ~11.9, 3.86 active coils)
14	1	F1S-401	Front hub / stub axle, Ti-6Al-4V, centre-lock nose, disc bell
15	1	F1S-402	Hub bearings 2x 7014 ACD/P4A angular contact back-to-back (70 x 110 x 20 each) + 14 mm spacer ring (envelope)
16	1	F1S-403	Hub-bearing retaining nut M66x1.5, 4340 steel





F1SUSPENSION - DOUBLE-WISHBONE PUSHROD FRONT-LEFT CORNER (FULL SIZE)
TRACK 1600 WHEELBASE 3600 TYRE 30/68-18 GT3 SLICK (680 OD) ON 18X12 STATIC CAMBER -3 DEG
KPI 8.5 DEG CASTER 6.9 DEG SCRUB 41 TRAIL 29
TRAVEL +35/-25 MM CAMBER GAIN -0.54 DEG/25 MM RC 66 MM ANTI-DIVE 26 %
INSTALLATION RATIO 0.894 SPRING 158 N/MM WHEEL RATE 131 N/MM RIDE 4.14 HZ
ASSEMBLY PARAMETERS: Heave (WHEEL TRAVEL) AND Steer (RACK) DRIVE THE MECHANISM
EXPLODED VIEW = F1Suspension_Exploded.iam. FASTENERS ISO 4762 12.9, THREADS COSMETIC.
CHASSIS SECTION AND RACK HOUSING ARE REFERENCE GEOMETRY (NOT IN PARTS LIST). GROUND PLANE Z = 20.0 (680 MM TYRE ON THE 720 MM HARDPOINT LAYOUT)
DAMPER PENSKE 8760 (BUILT TO LENGTH) + HYPERCO 186A0900, DISC AP CP3870-129CR5 320 X 32, HUB 2X 7014 + SPACER

DRAWN Jay Patel	11/1/2026	
CHECKED		TITLE: F1Suspension - general assembly
QA		
MFG		
APPROVED		
		SIZE D · DWG NO F1S-DWG-001 · REV
		SCALE 1 / 8 · SHEET 1 OF 1